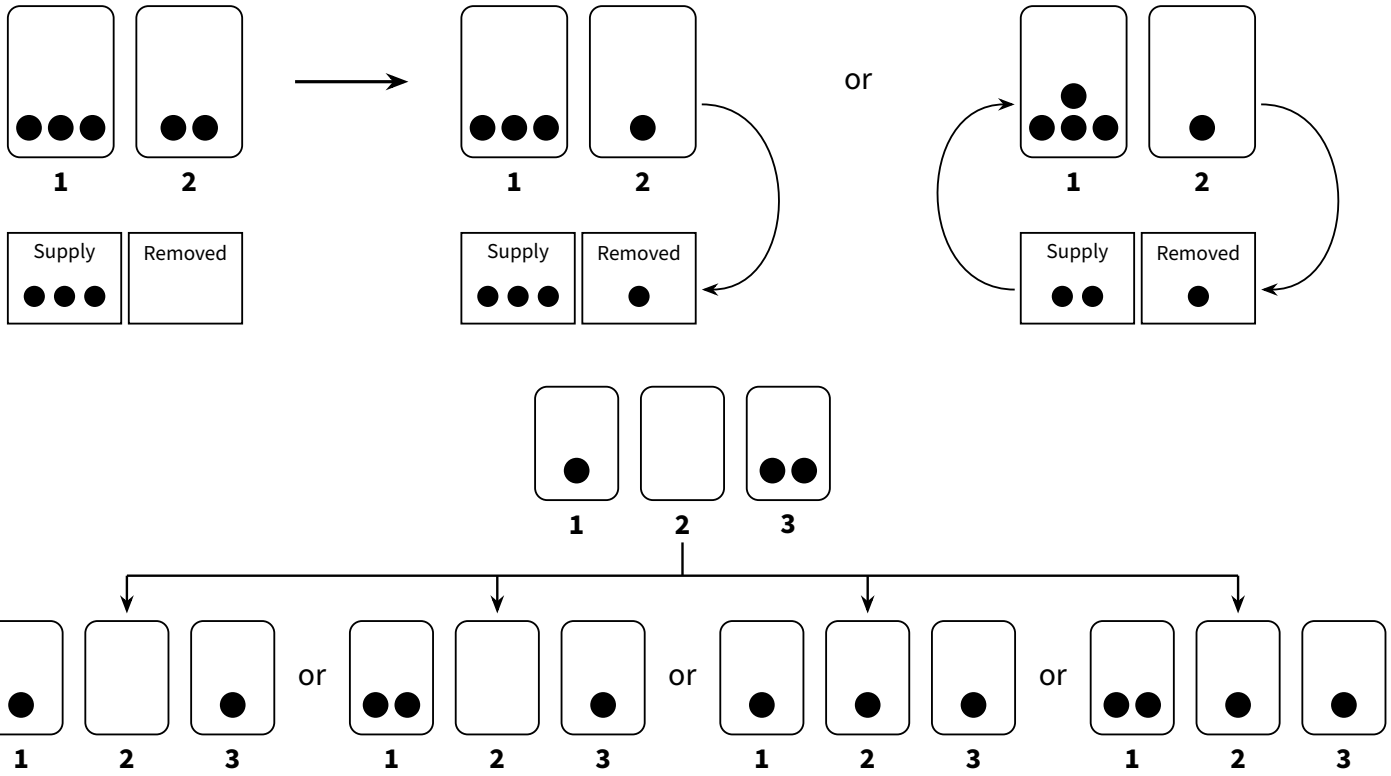
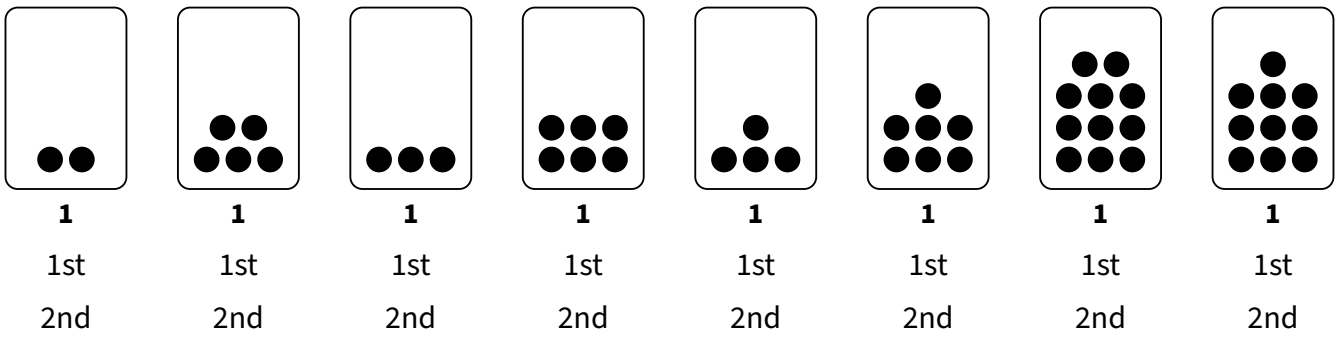
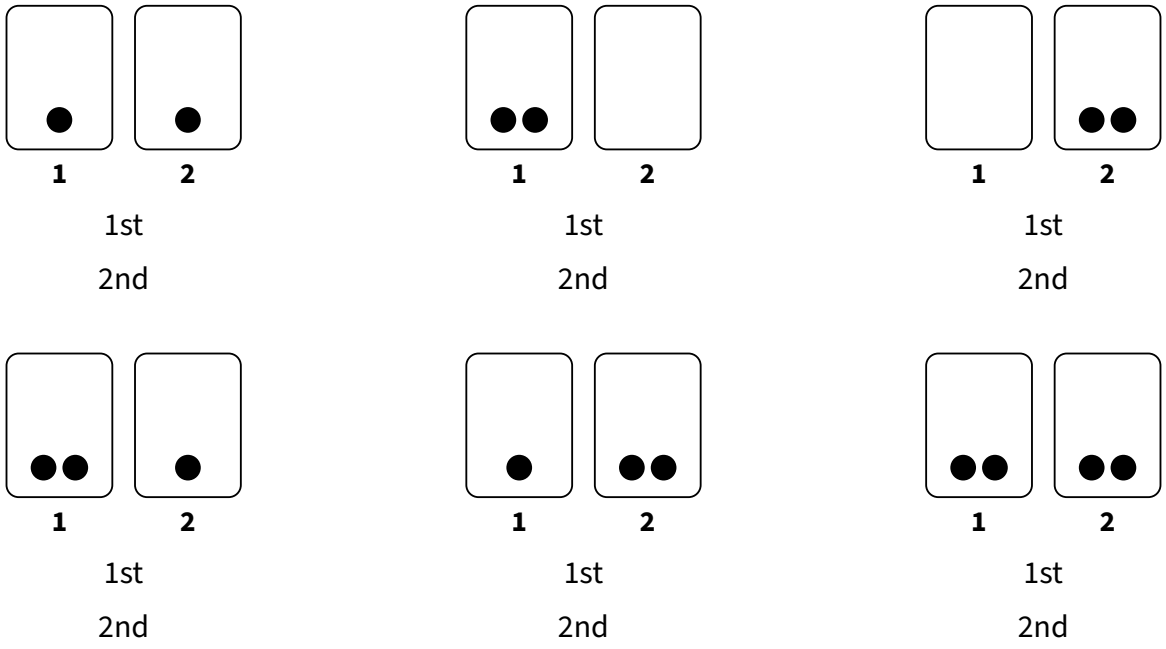
Two players take turns. On your turn, take one counter out of a bucket and put it in Removed. If you took it from bucket 2, you may also put one counter from Supply into bucket 1. If you took it from bucket 3, you may also put one counter from Supply into bucket 1, into bucket 2, or into both. Whoever takes the last counter wins. Start each game with Removed empty.



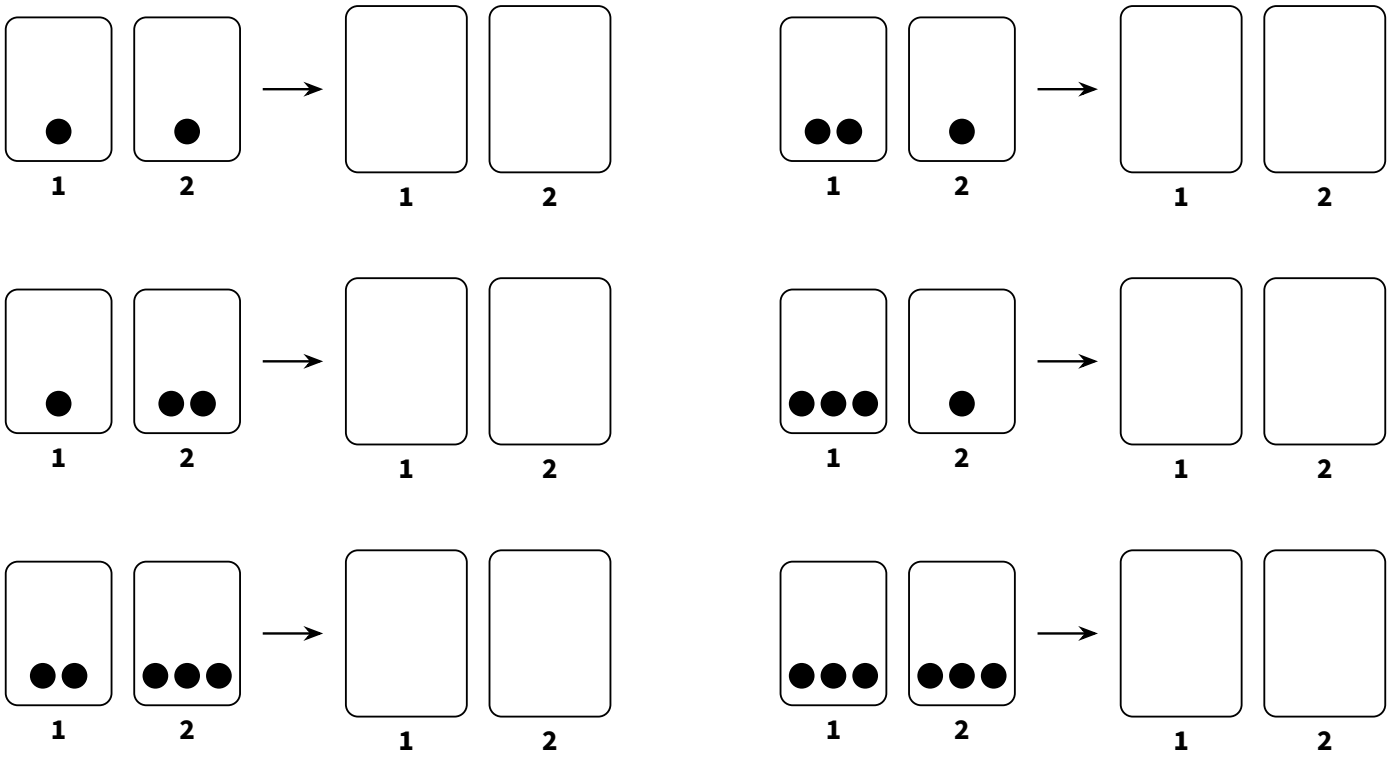
Problem 1: Cover buckets 2 and 3. Put counters in bucket 1 like each picture and play. Circle who wins.



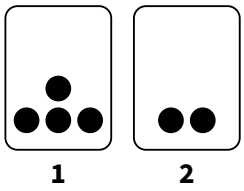
Problem 2: Uncover bucket 2. Play from each start, taking turns going first. Circle who can always win.



Problem 3: You go first from each start. Find a first move that lets you always win, and draw the buckets after it.

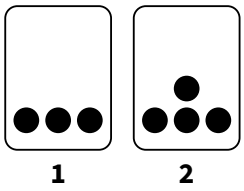


Problem 4: Play against the adult from each start, taking turns going first. Circle who can always win, and show the adult how.



1st

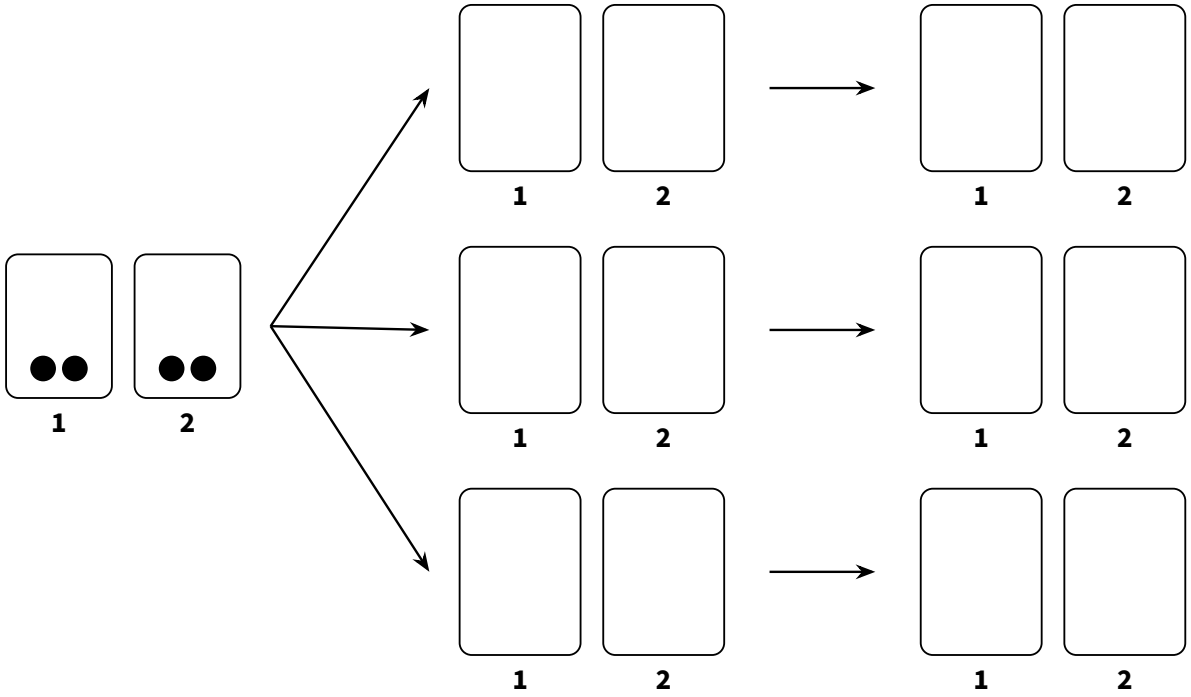
2nd



1st

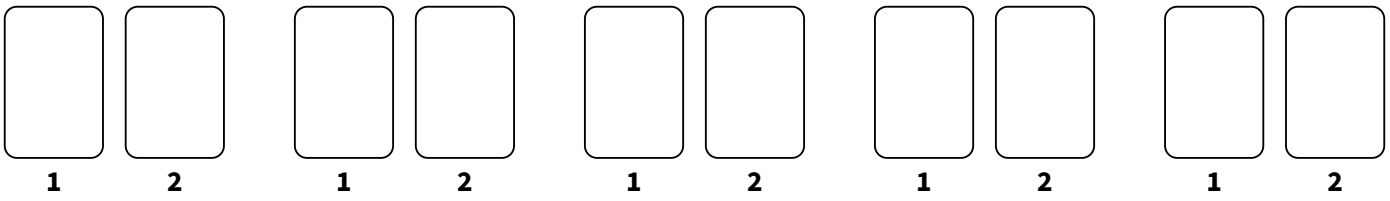
2nd

Problem 5: Your partner goes first from this start. Draw the buckets after each move your partner can make. Next to each, draw the buckets after a move that lets you always win.

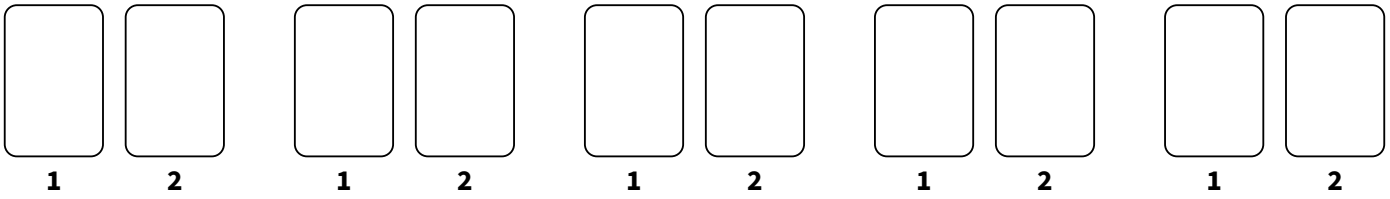


Problem 6: Find every way to put 4 counters into buckets 1 and 2 so that the player who goes second can always win. A bucket can be empty. Then do the same with 6 counters.

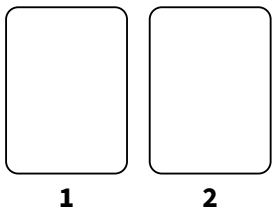
4 counters



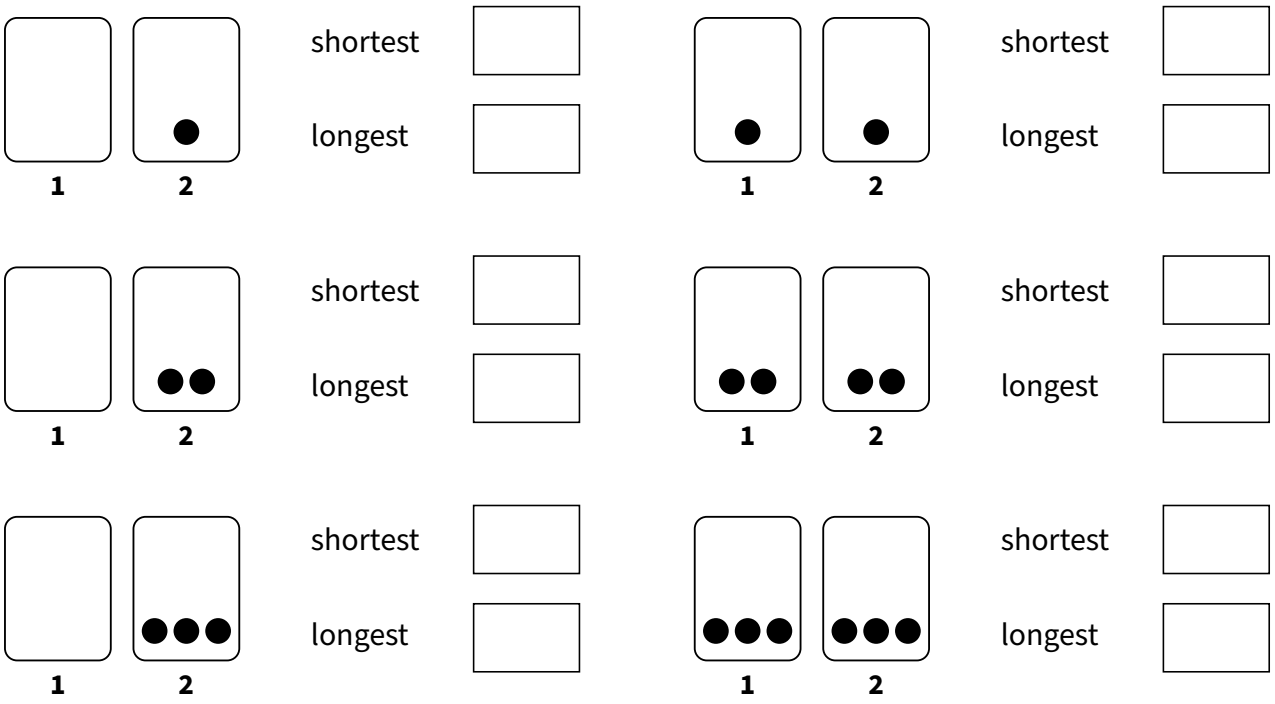
6 counters



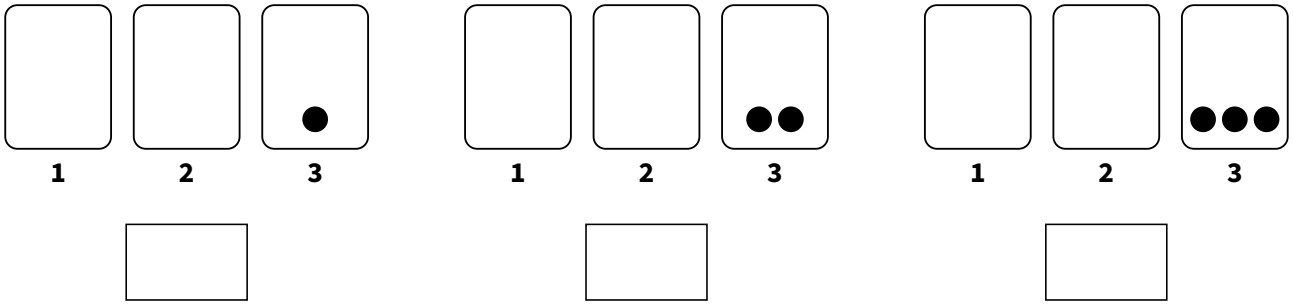
Problem 7: If you can, put 5 counters into buckets 1 and 2 so that the player who goes second can always win. If you can't, explain why.



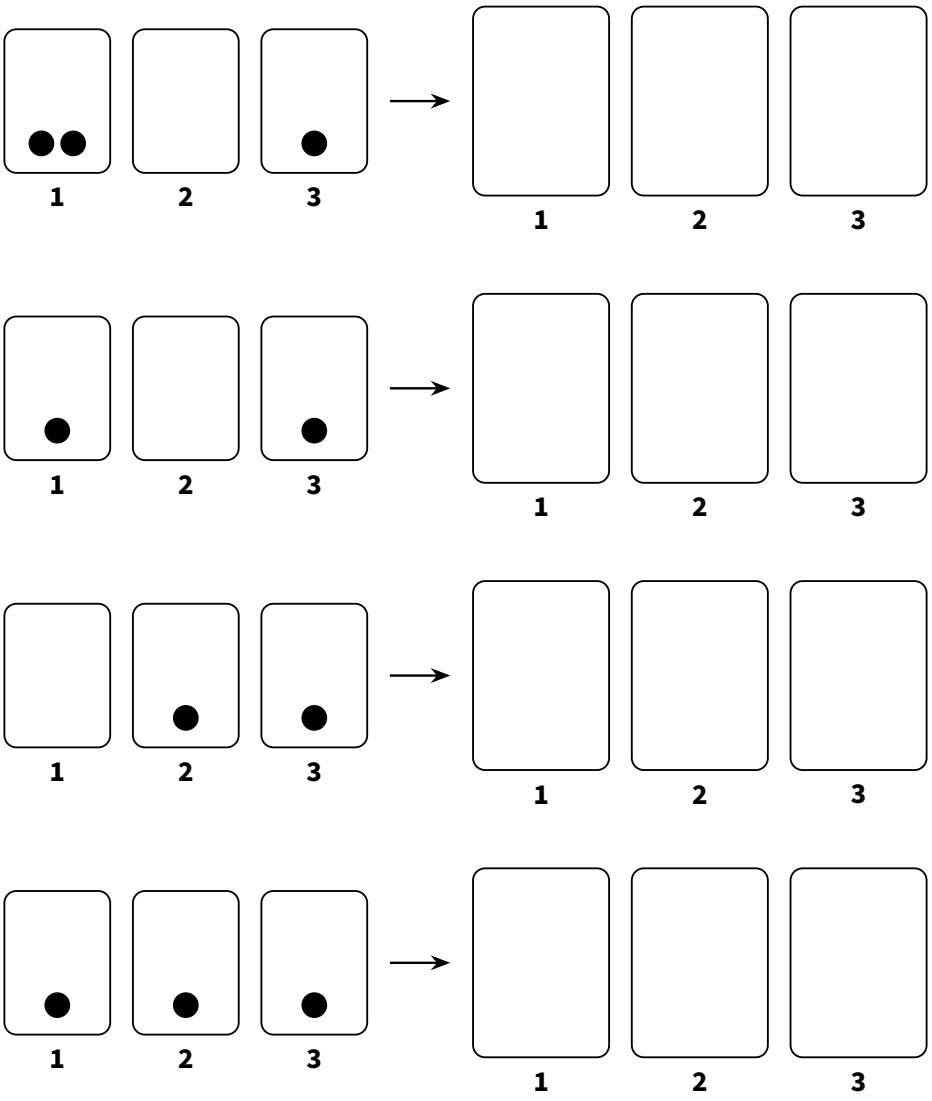
Problem 8: Work together. From each start, make the game as short as you can and then as long as you can. Write how many counters end up in Removed.



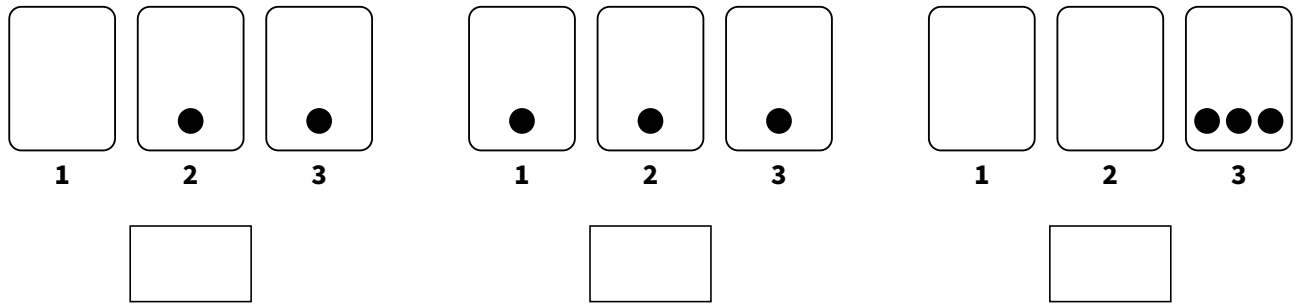
Problem 9: Uncover bucket 3. Work together to make the buckets hold as many counters as you can. Write the most they hold.



Problem 10: You go first from each start. Find a first move that lets you always win, and draw the buckets after it.



Problem 11: Work together to make the game from each start last as long as you can. Write how many counters end up in Removed.



Problem 12: Could a game from this start go on forever? Explain.

